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Department of Mathematics, Bioinformatics and Computer Applications
Mathematics-IV (MTH-241)

Assignment-3

Even Sem: 2024-25

Q.1. Solve the assignment problem represented by the following matrix.

	I	II	III	IV	V	VI
A	12	25	61	14	21	30
B	46	81	75	53	66	51
C	44	31	94	40	48	36
D	77	45	30	52	42	35
E	39	14	60	25	28	21
F	6	59	56	34	20	31

Q.2. A company has a team of four salesmen and there are four districts, where the company wants to start its business. After taking into account, the capabilities of salesman and the nature of districts, the company estimates that the profit per day in rupees for each salesman in each district is as follows

	D1	D2	D3	D4
A	20	14	18	15
B	18	15	19	19
C	19	19	17	16
D	17	16	18	19

Find the assignment of salesman to various districts which yields maximum profit.

Q.3. A company has four territories open, and four salesmen available for assignment. The territories are not equally rich in their sales potential and it is estimated that a typical salesman operating in each territory would bring in the following annual sales:

Territory	I	II	III	IV
Annual sales (Rs.)	60,000	50,000	40,000	30,000

The four salesmen are also considered to differ in ability and it is estimated that working under the same conditions their yearly sales would be proportionally as follows

Salesman	A	B	C	D
Proportion	7	5	5	4

If the criterion is maximum expected total sales, the intuitive answer is to assign the best salesman to the richest territory, the next best salesman to the second richest, and so on. Verify this answer by the assignment technique.

Q.4. An air-line that operates seven days a week has time table as shown below. Crews must have a minimum layover of 5 hours between flights. Obtain the pairing of flights that minimizes

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layover time away from home. For any given pairing, the crew will be based at the city that results in the smaller layover. For each pair, also mention the town where the crew should be based.

Flight No	Departure	Arrival
1	7A.M.	8A.M.
2	8A.M.	9A.M.
3	1.30P.M.	2.30P.M.
4	6.30P.M.	7.30P.M.

Table 1: Delhi-Jaipur

Flight No	Departure	Arrival
101	8A.M.	9.15A.M.
102	8.30A.M.	9.45A.M.
103	12Noon	1.15P.M.
104	5.30P.M.	6.45P.M.

Table 2: Jaipur-Delhi

Q.5. A company is faced with the problem of assigning six different machines to five different jobs. The costs are estimated as follows (hundreds of rupees). Solve the problem assuring that the objective is to minimize the total cost.

	1	2	3	4	5
1	2.5	5	1	6	1
2	2	5	1.5	7	3
3	3	6.5	2	8	3
4	3.5	7	2	9	4.5
5	4	7	3	9	6
6	6	9	5	10	6

Q.6. Solve the assignment problem represented by the following matrix

	I	II	III	IV
A	10	12	9	11
B	5	10	7	8
C	12	14	13	11
D	77	45	30	5
E	8	15	11	9

Q.7. A city corporation has decided to carry out road repairs on main four arteries of the city. The government has agreed to make special grant of Rs. 50 lakhs towards the cost with a condition that the repairs must be done at the lowest cost and quickest time. If conditions warrant, then a supplementary taken grant will also be considered favourably. The corporation has floated tenders and 5 contractors have sent in their bids. In order to expedite work, one road will be awarded to only one contractor. Find the best way of assigning the repair work to the contractors and the costs.

Contractors/ Cost of repair (Rs.)	R1	R2	R3	R4
C1	9	14	19	15
C2	7	17	20	19
C3	9	18	21	18
C4	10	12	18	19
C5	10	15	21	16

Q.8. A small garment making unit has five tailors stitching for five different types of garments. All the five tailors are capable of stitching all the five types of garments. The output per day per tailor and the profit (Rs.) for each type of garment are given below

Tailors\Garments	1	2	3	4	5
A	7	9	4	8	6
B	4	9	5	7	8
C	8	5	2	9	8
D	6	5	8	10	10
E	7	8	10	9	9
Profit(Rs.) per garment	2	3	2	3	4

- (i) Which type of garment should be assigned to which tailor in order to maximize the profit by assuming that there are no other constraints?
- (ii) If tailor D is absent for a specified period and no other substitute tailor is available, what should be the optimal assignment?

Q.9. The captain of a cricket team has to allot five middle batting positions to five batsmen. The average runs scored by each batsman at these positions are as follows

Batsman/Batting position	I	II	III	IV	V
P	40	40	35	25	50
Q	42	30	16	25	27
R	50	48	40	60	50
S	20	19	20	18	25
T	58	60	59	55	53

- (i) Find the assignment of batsmen to the positions which would give the maximum number of runs.
- (ii) If another batsman 'U' with the following average runs in batting positions as given below is added to the team, should he be included to play in the team? If so, who will be replaced by him?

Batting positions	I	II	III	IV	V
Average runs	45	52	38	50	49

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